

Stress before Distress

Households' Perceived Delinquency Risks and Realized Outcomes

Tao Wang

Bank of Canada

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The views expressed in this paper are those of the author and do not necessarily reflect those of the Bank of Canada.

Motivation

- Households' financial **stress** is a salient feature of modern capitalist societies
 - Gallup (2024): 28% (48%) of U.S. households worry about a missed (bill) payment
 - FCAC (2025): 56% of Canadians report trouble meeting commitments

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- A gap commonly emerges between **perceived stress** and **realized distress**
 - Often a source of policy debate: **widespread stress** does not match **actual arrears**
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 - Tempting to dismiss as mere **sentiment** or survey error, and track only **hard numbers**
- This paper, however, takes both measures seriously, and asks:
 - Why is **distress** rare while **stress** prevalent?
 - What links **stress** to **distress**?
 - Does this gap matter for the macroeconomy? (not today)

Preview

Part I – Empirics: how **stress** maps into **distress**

1. **Stress** reasonably co-moves with other expectations
2. **Stress** predicts **distress**, but does not always materialize as the latter
3. Empirical measure of the wedge attributable to macro shocks and self-insurance

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Part II – A life-cycle model of consumption, borrowing, and delinquency

1. Reconciles broad **ex-ante stress** with narrow **ex-post distress**
2. Model view of the wedge: forward-looking self-insurance – anticipating **stress**, households cut consumption, prepay debt, build buffers
 - **Stress**: subjective expectation of the outcome absent of self-insurance responses
 - **Distress**: realization of the outcome mitigated by self-insurance

Related literature

- **HA-macro with credit, bankruptcy, and financial stress.** Faccini et al. (2026), Sergeyev, Lian, and Gorodnichenko (2023), Auclert, Dobbie, and Goldsmith-Pinkham (2019), Athreya, Mather, et al. (2019), and Mitman (2016)
→ We focus on unsecured debt *delinquency* risks and their behavioral responses
- **Equilibrium models of consumer default & delinquency.** Athreya, Sanchez, et al. (2018), Athreya, Sánchez, et al. (2015), Chatterjee et al. (2007), and Livshits, MacGee, and Tertilt (2007)
→ We use a partial-equilibrium model to explain the subjective financial *stress*
- **Subjective expectations in structural models.** Koşar and O'Dea (2023)
→ We show subjective *stress* is not model-consistent belief
- **Expectations in consumer credit.** Exler et al. (2024) and D'Acunto, Weber, and Yin (2024)
→ we study *perceived delinquency risk* and its link to other macro expectations

Data: measuring stress and distress

- **Stress** — *perceived* probability of missing a minimum debt payment over the next 3 months
 - U.S.: NY Fed **Survey of Consumer Expectations** (SCE), monthly, 2013–2024
 - Canada: Bank of Canada **CSCE**, quarterly, 2014–2024 — identical question
- **Distress** — *realized* delinquency rates from credit registries
 - U.S.: FRBNY **Consumer Credit Panel** (Equifax, 5% panel)
 - Canada: Bank of Canada anonymized **TransUnion** microdata
- Analysis at the **population level**: each measure averaged across households
- Supplementary: **SCF** 2019 (U.S. life cycle), FCAC **Well-Being Monitor** (Canada)

Outline

Empirics: Stress and Distress

Perceived risk as financial stress

Does stress realize?

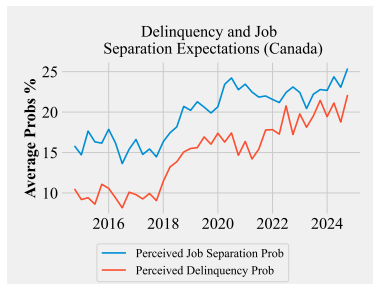
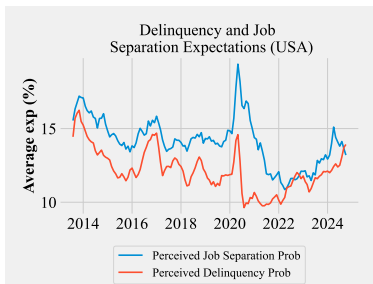
From stress to distress

A Life-Cycle Model of Delinquency

Conclusion

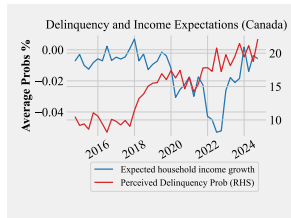
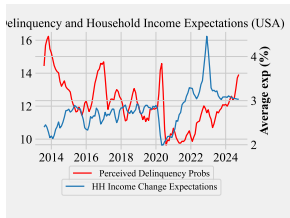
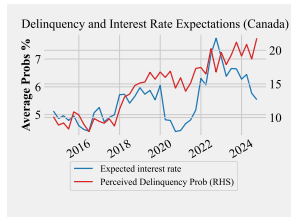
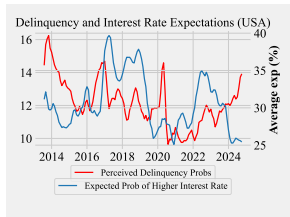
Stress co-moves with unemployment risk

- **Perceived** delinquency risk co-moves with **perceived** job-loss risk: U.S. $\rho = 0.57$, Canada $\rho = 0.43$
- **Realized** job flows track **realized** delinquency too (pandemic transfers broke the link) realized job flows



Left: U.S. (SCE, monthly). Right: Canada (CSCE, quarterly).

...and with rates, income, and debt burden

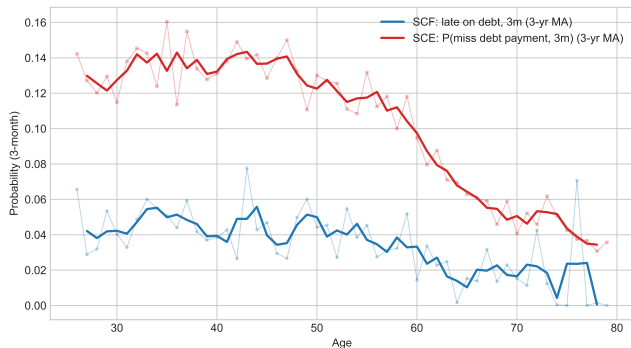


- Canada: **perceived risk** ↓ with expected income growth ($\rho \approx -0.50$), and ↑ with expected borrowing-rate hikes ($\rho \approx 0.68$)
- U.S.: income $\rho \approx -0.30$; interest $\rho \approx 0.05$ (SCE asks the *saving* rate)
- **DSI**: **perceived risk** co-moves with aggregate indebtedness (U.S. 0.48, Canada 0.38) DSI plots

⇒ a credible proxy for **financial stress**.

Top: rate exp.; bottom: income exp. Left: U.S.; right: Canada.

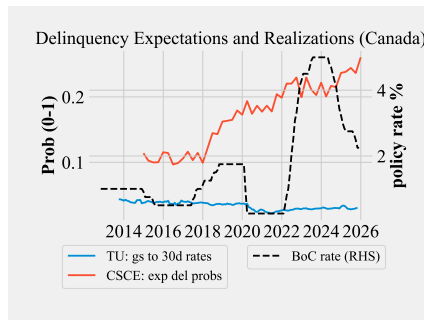
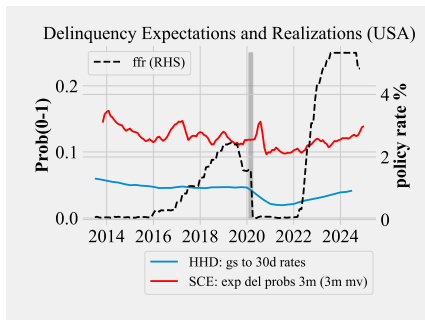
Over the life cycle: perceived $\approx 3\times$ realized



- **Realized delinquency** (SCF 2019):
~4–5% young, falls below 2% by late 60s
- **Perceived delinquency risk** (SCE):
peaks ~14% in the late 30s–40s, declines to $< 5\%$
- Both decline monotonically with age, by roughly the same proportion

Does stress realize? U.S. yes, Canada no

- Ex-ante **perceptions** vs. ex-post **transition rates**: $\rho = 0.70$ (U.S.) but -0.70 (Canada)
- ⇒ **Perceived** sits *above* **realized** in both: **stress** (broad) vs. **distress** (narrow)



Left: U.S. (monthly). Right: Canada (quarterly). Source: FRBNY CCP, TransUnion, SCE/CSCE.

The Stress-to-Distress Ratio (SDR)

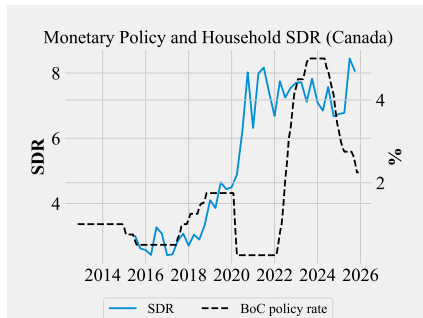
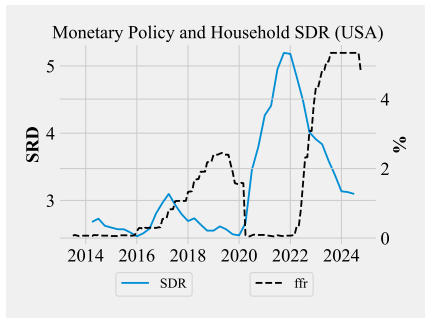
Realized delinquency = perceived probability scaled by insurance and hit by unexpected shocks:

$$\log\left(\frac{\widetilde{\text{Prob}}(D_{i,t+1} \mid \Omega_{i,t})}{\text{Prob}(D_{i,t+1})}\right) = \underbrace{-(\psi_{t+1} + \phi)}_{\text{SDR}} \frac{\text{stress (perceived)}}{\text{distress (realized)}}$$

- **SDR** = combined effect of unexpected macro shocks ψ and insurance ϕ (self- and public)
- Higher SDR $\Rightarrow$ a smaller portion of perceived risk materializes as realized delinquency $\Leftarrow$ more effective insurance or less adverse macro realization

Macro driver: monetary policy rate $\uparrow \rightarrow$ SDR $\downarrow$ (U.S.)

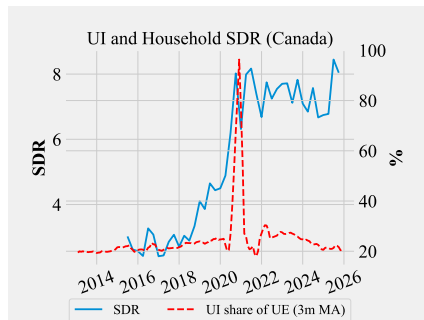
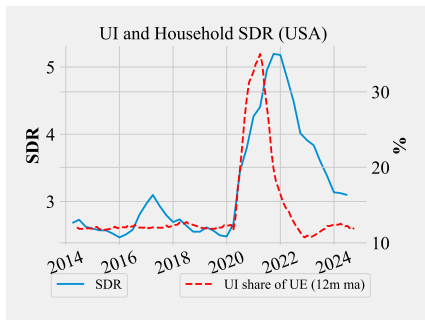
- U.S.: tightening (loosening) is followed by an almost synchronous *fall* (rise) in the SDR — it cycles with policy
- Canada: SDR shows a single *upward trend*, largely insensitive to policy cycles



Left: U.S. (monthly). Right: Canada (quarterly). Policy rate on right axis; SDR in exponential form.

Macro driver: more generous UI $\rightarrow$ SDR $\uparrow$ (both)

- SDR is strongly correlated with the **UI reciprocity rate** – the share of the unemployed receiving benefits – in both countries
- **Public insurance** insulates the **stress** $\rightarrow$ **distress** link



Left: U.S. (monthly). Right: Canada (quarterly). Right axis: UI reciprocity rate.

Self-insurance: higher stress → lower current spending

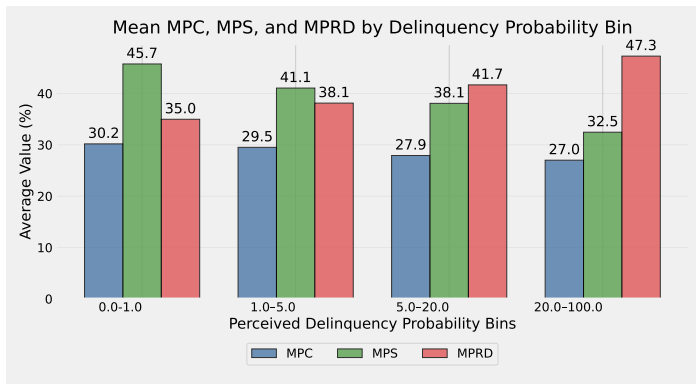
- Higher **perceived** delinquency risk → higher **expected** spending growth:
households plan to **cut spending today relative to tomorrow**

	(1)	(2)	(3)	(4)
Job Separation Probs			0.03*** (0.00)	
Job Finding Probs				0.00** (0.00)
Delinquency Probs	0.04*** (0.00)	0.03*** (0.00)	0.02*** (0.00)	0.04*** (0.00)
Expected Wage Growth	0.01 (0.01)			
Perceived Wage Risks		0.01*** (0.00)		
<i>N</i>	20191	20191	20188	20189
<i>R</i> ²	0.02	0.02	0.03	0.02

Dep. var.: expected spending growth (Canada CSCE). SE clustered by household; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Self-insurance: MPC ↓, MPRD ↑

- As **perceived delinquency risk** rises, households shift marginal resources from consumption toward **debt repayment** rather than saving



Taking stock: what the empirics tell us

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4. **Beliefs capture exposure, not the cure.** **Stress** reflects *risk exposure*, but not the insurance that contains **distress**

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⇒ We formalized this interpretation using a life cycle consumption/borrowing/delinquency model.

Outline

Empirics: Stress and Distress

A Life-Cycle Model of Delinquency

- Setup and calibration

- Micro behaviors

- Aggregate dynamics

Conclusion

Key Ingredients

- **Income risk:** life cycle profile + permanent/transitory wage shocks + U–E transitions
- **Expenditure shocks:** reduce resources (medical, mortgage rate resets, etc.)
- **Preference shocks:** unobserved shocks influencing pay/not-pay
- **Delinquency costs:** penalty interest, exclusion risk, utility drop
- **Rate differential:** borrowing rate $>$ saving rate $\Rightarrow$ kink at zero wealth
- **Borrowing constraint:** exogenous, varies by age
- **Subjective beliefs:** households act on perceived income/employment risk that differs from realized (Wang 2023)

Model: Environment

- **Preferences** – CRRA flow utility + warm-glow bequest:

$$\mathbb{E}_0 \left[\sum_{t=0}^L \beta^t u(c_t) + \beta^{L+1} \nu u(s_{L+1}) \right], \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

- **Single net balance** s with kinked rates and an exogenous borrowing limit:

$$\mathcal{R}(s) = \begin{cases} R_s & s \geq 0 \\ R & s < 0 \end{cases} \quad (R > R_s), \quad s' \geq -\chi$$

- **Income** $y = \Gamma_t e^p e^\theta W_t$: life-cycle profile Γ_t , permanent $p_t = p_{t-1} + \psi_t$, transitory θ_t , employment x (Markov); **economy-wide wage** W_t
- **Beliefs**: continuation values use **perceived** risks $\tilde{\mathbb{E}}$; the simulation draws from the **actual** risks
- **Delinquency costs**: penalty rate $R(1 + \eta)$, re-entry prob. λ , flow utility $-\delta$

Model: Recursive Problem

Observe state $(s, p, \theta, \kappa, x)$ and taste shocks; choose *pay/not pay* then c . Primes = next period; $\tilde{\mathbb{E}}$ = perceived.

$$\textbf{Pay / save: } v^n(s, \cdot) = \max_{c \geq 0} u(c) + \beta \tilde{\mathbb{E}} W'$$

$$\text{s.t. } y + s \mathcal{R}(s) - \kappa = c + s', \quad s' \geq -\chi$$

$$\textbf{Delinquent: } v^d(s, \cdot) = u(y - \kappa) - \delta + \beta \tilde{\mathbb{E}}[(1 - \lambda) v^{d'} + \lambda W']$$

$$\text{s.t. } s' = s R(1 + \eta), \quad s' \geq -\chi$$

Discrete choice (taste shocks) $\Rightarrow$ log-sum: $W = \sigma \log[e^{v^n/\sigma} + e^{v^d/\sigma}]$

Savers ($s \geq 0$) always pay; *forced* delinquency when paying is infeasible.

Solution: upper-envelope EGM of Iskhakov et al. (2017).

Calibration

Preferences & demographics

β	discount factor	0.94
γ	CRRA / bequest curvature	2
ν	bequest scale	1.0
L_R, L	retirement / terminal age	65, 85

Prices & credit

R_s, R	gross save / borrow rate	1.02, 1.08
χ	borrowing limit (/ perm. inc.)	1
W	wage rate (normalized)	1.0

Delinquency

η	penalty markup on R	0.02
λ	re-entry probability	0.90
δ	flow utility penalty	0.7
σ	taste-shock scale	0.8

Income risk — actual vs. perceived

	actual	perceived
σ_ψ permanent SD	0.10	0.03
σ_θ transitory SD	0.10	0.03
π_{ue} job-finding	0.82	0.77
π_{eu} job-loss	0.04	0.04

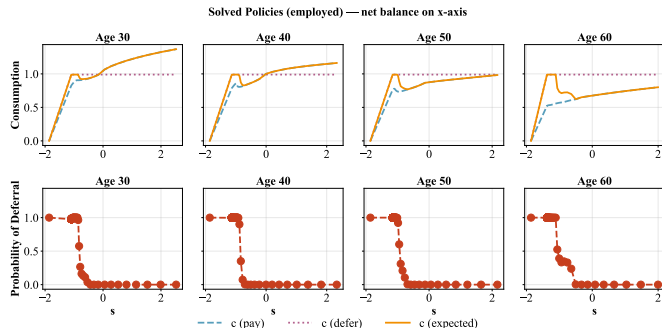
Insurance & expenditure

ζ	UI replacement ratio	0.4
ρ^{SS}	SS replacement ratio	0.6
κ	expend. shock $\sim \Gamma(\alpha, \beta)$	(0.02, 0.10)

Γ_t : hump-shaped life-cycle profile . Taxes

$\tau = \tau_{SS} = 0$; levels normalized by permanent income.

When to pay and when not to pay



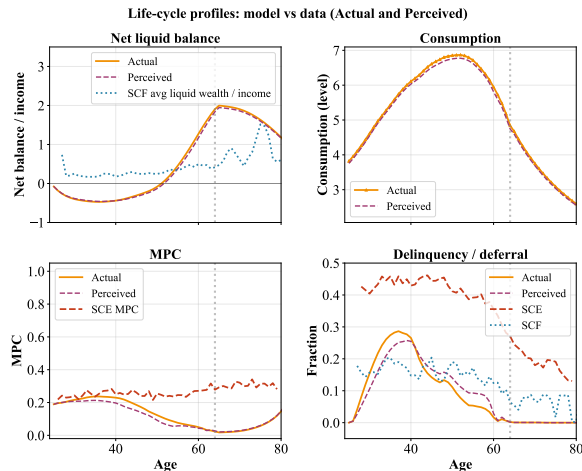
- *Near constraint*: high delinquency probability; *away*: none

- Region shifts with age and income shocks

⇒ Delinquency prob. is a continuous, forward-looking stress measure at every state

- Net balance s alone is a coarse proxy

Life Cycle Profiles



- Hump-shaped delinquency, concentrated at **young working ages**
- **Actual** (solid): optimize under perceived risk, realized outcomes from the true process
- **Perceived** (dashed): subjective beliefs treated as the truth

Micro Behaviors

Micro-Level Implications

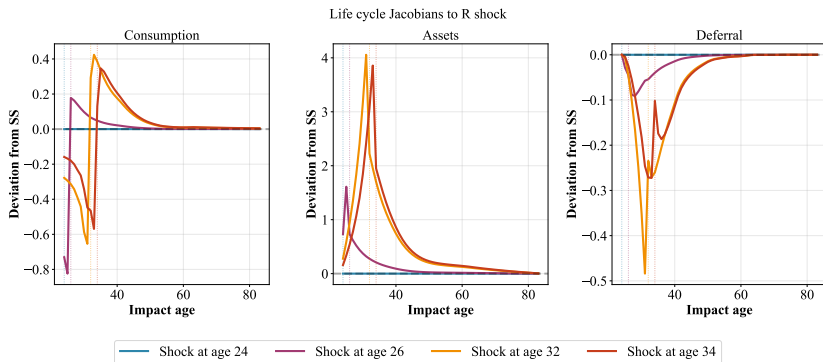
Four implications of the delinquency margin:

- **Non-monotonic MPCs:** the MPC depends on repayment probabilities, unlike the monotonic buffer-stock benchmark [details](#)
- **Sharp rise in delinquency near the constraint:** flat at low utilization, hockey-stick rise near the limit [details](#)
- **Precautionary prepayment:** penalty scales with debt $\Rightarrow$ accelerate repayment as self-insurance [details](#)
- **Voluntary vs. involuntary delinquency:** most — but not all — delinquency is involuntary [details](#)

Life-cycle Dynamics via the Jacobian

- **Life-cycle Jacobian:** cohort analogue of the sequence-space Jacobian (Auclert et al. 2021)
- $L \times L$ matrix: response of the cohort path $\{C_j, A_j, D_j\}$ to an anticipated R or W shock at each age
- **Anticipation** reshapes behavior *backward*; **impact & propagation** carry it *forward*
- We report: (i) the response to a one-time shock, and (ii) cross-age heterogeneity

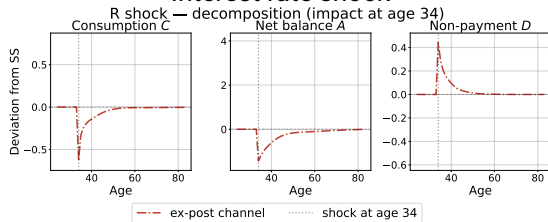
Life-Cycle Response: Interest Rate (R) Shock



- Anticipating the hike: **cut consumption, pay down debt**, non-payment falls; reverses at the shock age
- Mirror image for an income/wage gain (► W shock)

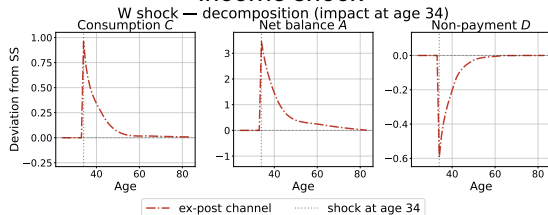
Jacobian Decomposition at a Given Age (Age 34)

Interest rate shock



- **Stress** stems only from the **ex-post channel**: nonzero only from the shock age forward

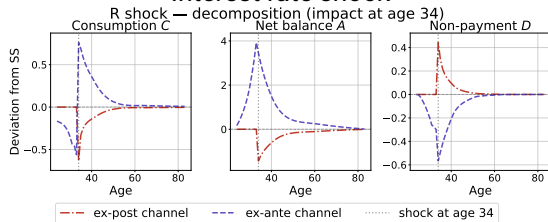
Income shock



► belief mitigation

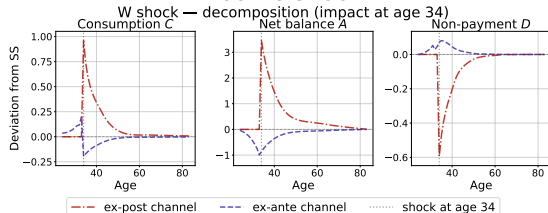
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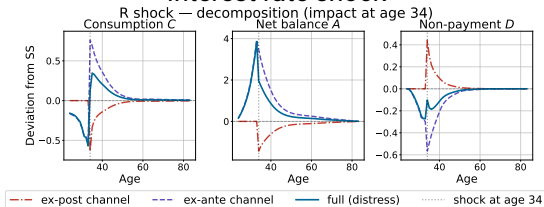
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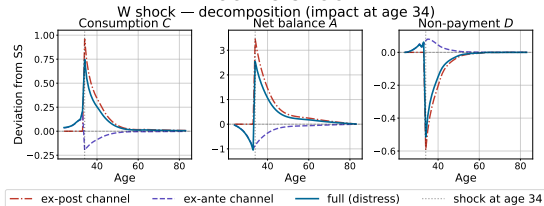
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- **Distress** combines both channels

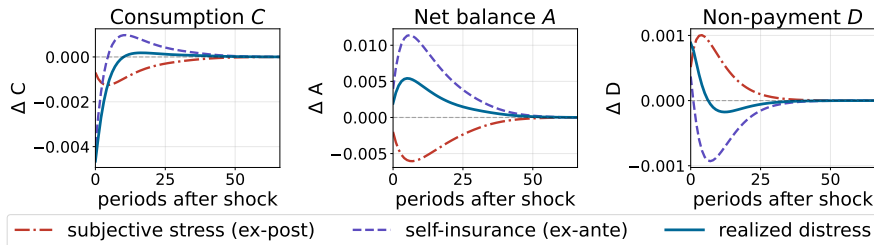
$$\Rightarrow \text{SDR} \approx \text{ex-post} / \text{full impacts}$$

► belief mitigation

Aggregate Dynamics

Impulse Responses to Persistent Aggregate Shocks

R shock — IRF (one s.d., $\rho=0.85$)



- **Stress** = pure ex post impacts of the shock absent of the self-insurance;
- **Distress** contains the self-insurance under subjective beliefs

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- **Perceived risk** is a **forward-looking** gauge of **stress**: co-moves with job, rate, income, debt risks; falls with age
- **Stress**→**distress** tracks the cycle, buffered by **public** and **self-insurance** and milder-than-feared shocks

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- **Stress**→**distress** tracks the cycle, buffered by **public** and **self-insurance** and milder-than-feared shocks
- A life-cycle model with a **delinquency choice** delivers a *continuous* model-implied **stress** measure, mapped directly to the data
- This model reconciles the **stress**–**distress** wedge arising from **self-insurance** and **subjective beliefs**

Why a life-cycle model of delinquency?

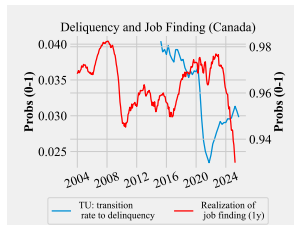
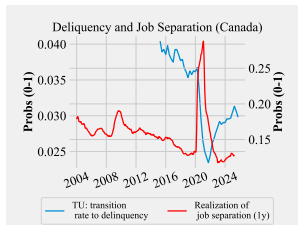
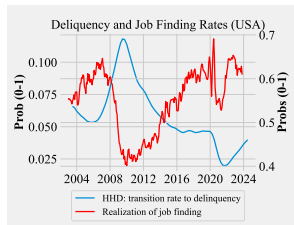
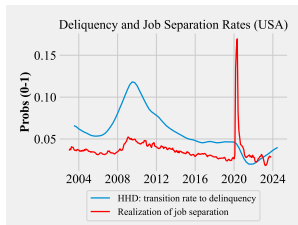
The life-cycle model of consumption, saving/borrowing, and **delinquency** contain following characteristics:

- **Delinquency as a “speed limit,” not a crash:** costly, but not too costly
→ Existing models jump from repayment to default; this framework fills in between
- **Coping capacity under constraints:** delinquency as a margin of adjustment when resources are tight
- **Forward-lookingness:** anticipating **stress**, households act — adjust consumption, prepay debt, build buffers (**self-insurance**)

Limitations / What the Model Does Not Do

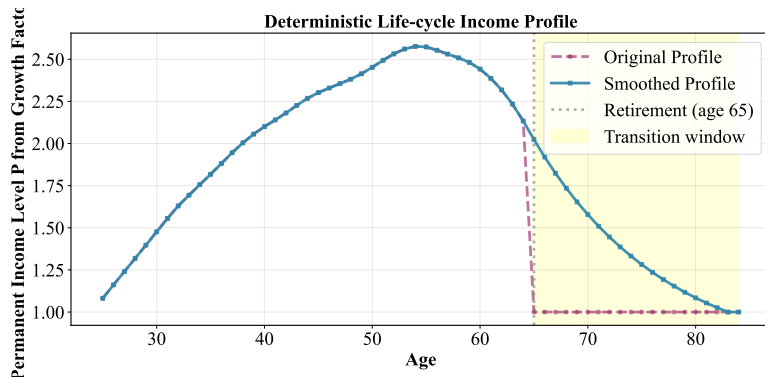
- **No mortgage / secured debt default** — unsecured credit only
 - *But:* can still think about a mortgager's problem facing payment shocks
- **No persistent credit score impacts** — exclusion risk is stylized
 - *But:* the utility cost captures it somewhat
- **Single asset** — no portfolio choice or liquid/illiquid split
 - *But:* implicitly assumes illiquid asset adjustment is costly
- **No co-holding** — no simultaneous saving and borrowing
 - *But:* still captures revolving behaviors and non-monotonic MPCs
- **No costly bankruptcy** — no foreclosure, fresh start, or permanent exclusion
 - *But:* captured by the hard borrowing constraint

Realized job flows track realized delinquency



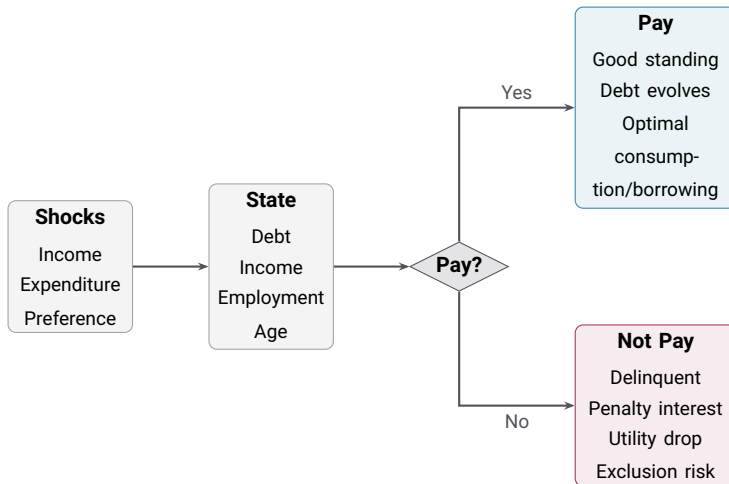
Realized job-separation (left) & job-finding (right) vs. realized delinquency. Top: U.S., bottom: Canada.

Deterministic Life Cycle Income Profile



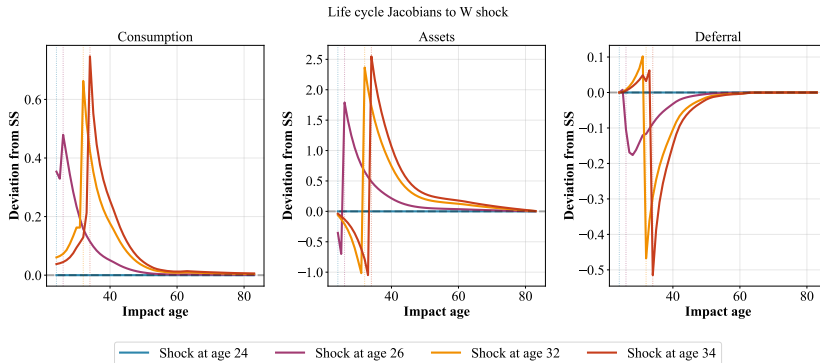
- Hump-shaped profile: peaks mid-50s, declines into retirement
- Smoothed transition around retirement (age 65) for numerical stability

Model at a Glance



Repeats over the life cycle: young → middle → old

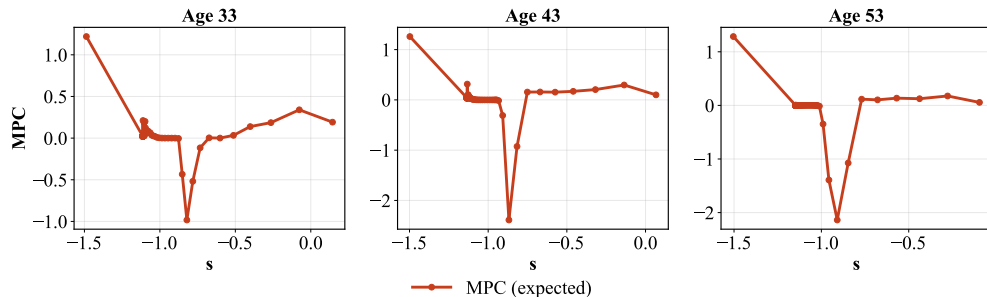
Life-Cycle Response: Income/Wage (W) Shock



- Mirror image: anticipating the gain, households **borrow ahead**, non-payment rises first, then subsides

Implication: Non-Monotonic MPCs

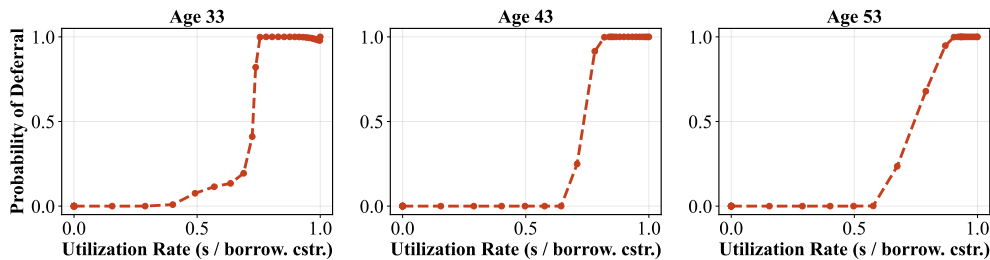
Marginal Propensity to Consume (employed) — net balance on x-axis



- MPC depends on repayment probabilities
- Contrast: buffer-stock model $\Rightarrow$ monotonic MPC

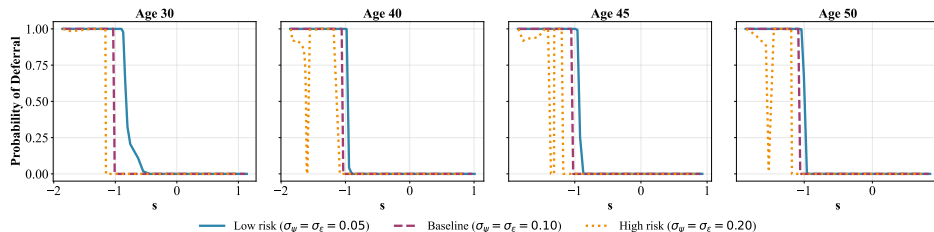
Implication: Sharp Rise in Delinquency near the Constraint

Deferral Probability vs Credit Utilization (employed)



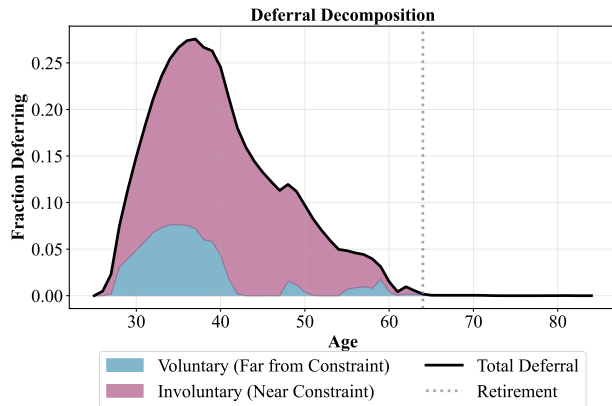
- Flat at low utilization, hockey-stick rise near limit
- Low utilization: has slack absorbs shocks, but can still fall into delinquency

Implication: Precautionary Prepayment



- Penalty scales with debt $\Rightarrow$ higher expected risk $\Rightarrow$ **accelerate repayment**
- Self-insurance, distinct from precautionary *saving*; stronger with κ /unemployment risk

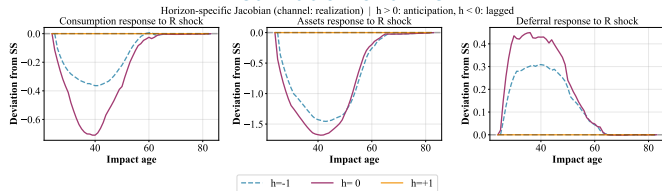
Implication: Voluntary vs. Involuntary Delinquency



- **Voluntary:** could repay, chooses not to, “splurge” consumption
- **Involuntary:** cannot repay – insufficient resources
- **Key result:** most but not all delinquency is involuntary

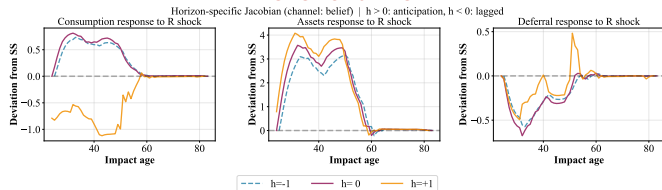
Heterogeneous Responses across Horizons: R Shock

Realization channel



- **Realization** channel is zero before the shock ($h = +1$); **belief** channel is active at all horizons
- **Belief** response **flips sign** for past vs. future shocks – precaution builds, then unwinds

Belief channel

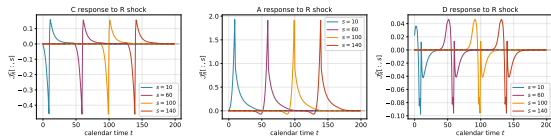


◀ back

From Life-Cycle to Aggregate Jacobian

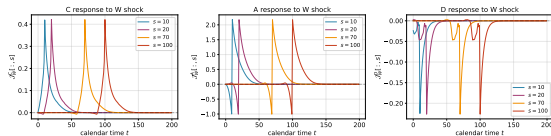
R shock

Aggregate Jacobian columns at several shock times



W shock

Aggregate Jacobian columns at several shock times



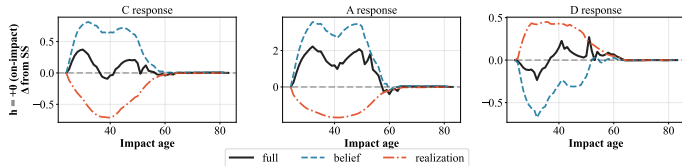
- Each column = population-weighted average of the life-cycle Jacobian (cohort weights from the stationary age distribution)
- The belief channel populates the off-diagonal mass — anticipated shocks bite away from impact

◀ back

Belief Channel Mitigates Stress Impact of Shocks

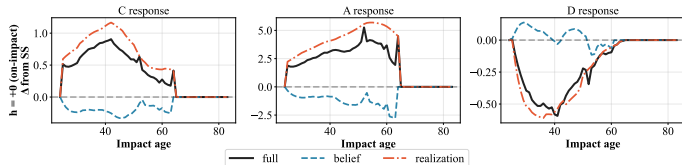
Interest rate shock

R shock — horizon $\times$ channel breakdown



Income shock

W shock — horizon $\times$ channel breakdown

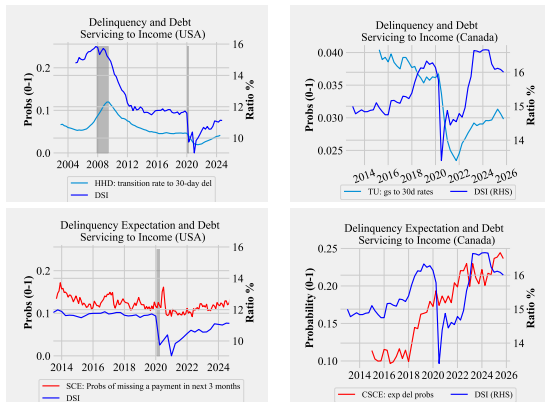


- Anticipation **absorbs** impact-period **stress**; total consumption/delinquency responses can flip sign at impact
- **Stress-to-distress** ratio differs across ages; can be both **positive** and **negative**

► across horizons

◀ back

Perceived risk co-moves with the debt-service ratio (DSI)



Bottom: **perceived** delinquency vs. DSI (ρ : U.S. 0.48, Canada 0.38). Top: **realized** delinquency vs. DSI (ρ : U.S. 0.77, Canada 0.05). Left: U.S.; right: Canada.

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